# Performance Evaluation of Multi-Objective Techniques for ZDT Function:

## A Comparative Study

*Comparative Analysis of NSGA-II and SPEA-II*


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# Abstract

This technical report presents a comprehensive performance evaluation and comparative study of two widely adopted multi-objective evolutionary algorithms: the Non-dominated Sorting Genetic Algorithm II (NSGA-II) and the Strength Pareto Evolutionary Algorithm II (SPEA-II). The evaluation is conducted on the ZDT1 benchmark function, which is a well-established test problem in the multi-objective optimization community. The objective of this study is to analyze and compare the convergence behavior, diversity maintenance, computational efficiency, and quality of the Pareto-optimal front approximations produced by both algorithms.

The ZDT1 function presents a challenging yet tractable test case with a convex Pareto front, making it suitable for assessing algorithm performance in a controlled and reproducible setting. Both algorithms were implemented with standard parameterizations and tested across multiple independent runs to ensure statistical rigor. Performance metrics including Generational Distance (GD), Inverted Generational Distance (IGD), Hypervolume (HV), and Spread are used for evaluation.

Results indicate that NSGA-II generally exhibits faster convergence and better spread of solutions along the Pareto front, while SPEA-II demonstrates competitive performance in terms of solution quality for certain run configurations. The findings of this study provide practical insights for selecting appropriate multi-objective optimization techniques based on problem characteristics and computational constraints.

# 1. Introduction

Multi-objective optimization (MOO) is a branch of mathematical optimization dealing with problems that involve the simultaneous optimization of two or more conflicting objectives. Unlike single-objective problems, where a single optimal solution exists, multi-objective problems yield a set of trade-off solutions known as the Pareto-optimal front. No single solution in this set can be improved in one objective without degrading performance in at least one other objective.

Real-world engineering and scientific problems frequently exhibit multiple conflicting objectives. Examples include minimizing cost while maximizing performance, or reducing weight while improving structural integrity. The ability to explore and approximate the Pareto front efficiently is therefore of paramount practical importance. Evolutionary algorithms (EAs) have proven to be particularly well-suited for multi-objective optimization due to their population-based nature, which allows them to produce multiple Pareto-optimal solutions in a single run.

Among the many multi-objective evolutionary algorithms (MOEAs) proposed in the literature, NSGA-II (Deb et al., 2002) and SPEA-II (Zitzler et al., 2001) are considered landmark contributions that have stood the test of time. Both algorithms have been extensively studied, applied, and cited in thousands of academic works. This report systematically compares their performance on the ZDT1 benchmark function to provide an objective assessment of their relative strengths and weaknesses.

## 1.1 Motivation

The selection of an appropriate optimization algorithm is a critical step in solving multi-objective problems. Different algorithms exhibit varying behaviors depending on problem topology, dimensionality, and the shape of the Pareto front. A systematic comparative study helps practitioners make informed algorithmic choices. The ZDT1 function is chosen for this study due to its well-understood properties, convex Pareto front, and widespread use as a standard benchmark.

## 1.2 Objectives

- To implement NSGA-II and SPEA-II using standard configurations on the ZDT1 benchmark function.
- To evaluate algorithmic performance using established multi-objective performance metrics.
- To identify scenarios in which each algorithm outperforms the other.
- To provide recommendations for practitioners selecting MOEAs for continuous optimization tasks.

# 2. ZDT1 Benchmark Function

The ZDT test suite (Zitzler, Deb, and Thiele, 2000) comprises a set of scalable multi-objective test problems designed to evaluate multi-objective evolutionary algorithms. The ZDT1 function is the first and most fundamental problem in this suite. It is defined as a two-objective minimization problem with a convex Pareto-optimal front.

## 2.1 Mathematical Formulation

The ZDT1 function is defined for n decision variables x = (x1, x2, ..., xn) where each xi is in the range [0, 1]. The objective functions are:

```
f1(x) = x1
f2(x) = g(x) * h(f1, g)
g(x) = 1 + 9 * (sum(x2..xn)) / (n - 1)
h(f1, g) = 1 - sqrt(f1 / g)
```

The true Pareto-optimal front is characterized by g(x) = 1, which means all variables x2 through xn are equal to zero. Under this condition, the Pareto front is described by the equation f2 = 1 - sqrt(f1) for f1 in [0, 1], which forms a convex curve in the objective space.

## 2.2 Properties of ZDT1

ZDT1 is a standard benchmark with several important properties. The Pareto front is convex, providing a smooth curve that algorithms must approximate uniformly. The problem has a single Pareto-optimal region, eliminating local optima complications. The scalability of the number of decision variables allows for controlled complexity analysis. Standard implementations use n = 30 decision variables, though n = 10 or n = 100 are also common.

# 3. Algorithm Descriptions

## 3.1 NSGA-II (Non-dominated Sorting Genetic Algorithm II)

NSGA-II was proposed by Deb, Pratap, Agarwal, and Meyarivan in 2002 as an improved version of NSGA. It introduced three key innovations: a fast non-dominated sorting procedure, a crowding distance assignment mechanism, and an elitist selection strategy. These contributions addressed the main shortcomings of its predecessor and established NSGA-II as the benchmark for multi-objective evolutionary computation.

### 3.1.1 Fast Non-Dominated Sorting

The cornerstone of NSGA-II is its fast non-dominated sorting algorithm, which organizes the population into fronts F1, F2, ..., Fk, where F1 is the set of non-dominated solutions (the first Pareto front), F2 contains solutions dominated only by F1, and so forth. The algorithm achieves this in O(MN^2) time complexity, where M is the number of objectives and N is the population size. This was a significant improvement over the O(MN^3) complexity of earlier approaches.

### 3.1.2 Crowding Distance

To maintain diversity among solutions on the same non-dominated front, NSGA-II uses a crowding distance metric. The crowding distance estimates the density of solutions surrounding a particular solution in objective space. Solutions with a larger crowding distance are preferred as they represent less crowded regions, encouraging a well-spread Pareto front approximation. The crowding distance for a solution is computed as the average side length of the cuboid in objective space formed by its nearest neighbors.

### 3.1.3 NSGA-II Procedure

At each generation, NSGA-II generates an offspring population Qt of size N through binary tournament selection (based on rank and crowding distance), simulated binary crossover (SBX), and polynomial mutation. The combined population Rt = Pt ∪ Qt of size 2N is then sorted using fast non-dominated sorting. The new parent population Pt+1 is constructed by accepting fronts sequentially until N solutions are selected. If a front partially fits, solutions are chosen based on crowding distance.

## 3.2 SPEA-II (Strength Pareto Evolutionary Algorithm II)

SPEA-II was proposed by Zitzler, Laumanns, and Thiele in 2001 as a successor to SPEA. It employs an archive of fixed size to store non-dominated solutions found during the search. SPEA-II introduced an improved fitness assignment scheme, a fine-grained density estimation technique, and an enhanced archive truncation method that preserves boundary solutions.

### 3.2.1 Fitness Assignment

SPEA-II uses a strength-based fitness assignment. For each individual i, a raw fitness R(i) is calculated as the sum of the strengths S(j) of all individuals j that dominate i, where S(j) is the number of individuals dominated by j. A density estimation D(i) is added to the raw fitness to break ties. The density is computed using the k-th nearest neighbor method in objective space, where k = sqrt(N + N_bar) and N_bar is the archive size.

### 3.2.2 Environmental Selection

After fitness assignment, all non-dominated individuals from the combined population and archive are copied into the new archive. If the archive size exceeds the limit, a truncation operator based on nearest-neighbor distances is applied iteratively, removing the solution with the smallest distance to its nearest neighbor. If the archive is smaller than the limit, dominated solutions with the best fitness values are added.

### 3.2.3 SPEA-II Procedure

The SPEA-II algorithm begins by initializing both the main population P0 and an empty archive A0. At each generation t: fitness values are assigned to all individuals in Pt and At; environmental selection fills the new archive At+1; if termination criteria are met, the non-dominated solutions from At+1 are returned; otherwise, binary tournament selection (on fitness), SBX crossover, and polynomial mutation produce the next population Pt+1.

# 4. Implementation Details

Both NSGA-II and SPEA-II were implemented in Python using the DEAP (Distributed Evolutionary Algorithms in Python) library. All experiments were conducted under identical computational conditions to ensure fair comparison. The key implementation parameters are summarized in Table 1.

| Parameter | NSGA-II | SPEA-II |
|---|---|---|
| Population Size | 100 | 100 |
| Archive Size | N/A | 100 |
| Number of Generations | 250 | 250 |
| Number of Variables (n) | 30 | 30 |
| Crossover Operator | SBX (pc=0.9) | SBX (pc=0.9) |
| Crossover Distribution Index | 20 | 20 |
| Mutation Operator | Polynomial | Polynomial |
| Mutation Probability | 1/n = 0.033 | 1/n = 0.033 |
| Mutation Distribution Index | 20 | 20 |
| Selection | Binary Tournament | Binary Tournament |
| Independent Runs | 30 | 30 |

*Table 1: Implementation Parameters for NSGA-II and SPEA-II*

## 4.1 Variable Bounds and Representation

Each decision variable xi was encoded as a real number in the range [0, 1], consistent with the ZDT1 problem definition. A real-coded representation was used for both algorithms, leveraging SBX crossover and polynomial mutation which are well-suited for continuous search spaces. No discretization or binarization of variables was performed.

## 4.2 Initialization

Initial populations were generated using a uniform random distribution over the decision variable space [0, 1]^30. The same set of random seeds was used for both algorithms across the 30 independent runs to ensure reproducibility and fairness. The archive in SPEA-II was initialized as empty at the start of each run.

# 5. Performance Metrics

A robust comparative study requires multiple complementary performance metrics, as no single metric captures all aspects of algorithm performance. The following four metrics were used in this study:

## 5.1 Generational Distance (GD)

Generational Distance measures the average Euclidean distance from each solution in the approximation set to the nearest solution in the true Pareto front. A lower GD indicates better convergence. GD is defined as GD = (1/|A|) * sqrt(sum(d_i^2)), where d_i is the minimum distance from solution i in the approximation set A to the true Pareto front P*.

## 5.2 Inverted Generational Distance (IGD)

IGD is the converse of GD: it measures the average distance from each point in the true Pareto front to the nearest solution in the approximation set. IGD captures both convergence and diversity simultaneously and is considered more informative than GD alone. A lower IGD value indicates a better overall approximation of the Pareto front.

## 5.3 Hypervolume (HV)

The Hypervolume indicator measures the volume of the objective space dominated by the approximation set and bounded by a reference point. It is the only unary metric that is Pareto-compliant, meaning a higher HV always corresponds to at least one solution being closer to the true front. A higher HV value indicates better performance. For ZDT1, the reference point (1.1, 1.1) was used.

## 5.4 Spread (Delta)

The Spread metric (also called Delta or Diversity metric) measures the distribution uniformity of solutions along the Pareto front. It is defined based on the Euclidean distances between consecutive solutions and the extreme solutions. A spread value of 0 indicates a perfectly uniform distribution. Lower values of Spread correspond to better diversity of solutions.

# 6. Results and Analysis

This section presents the experimental results obtained from 30 independent runs of each algorithm on the ZDT1 function. Results are reported as mean values with standard deviations across all runs. Statistical significance was verified using the Wilcoxon rank-sum test at a significance level of 0.05.

## 6.1 Convergence Analysis

Table 2 presents the mean and standard deviation of the Generational Distance (GD) and Inverted Generational Distance (IGD) for both algorithms over 30 runs.

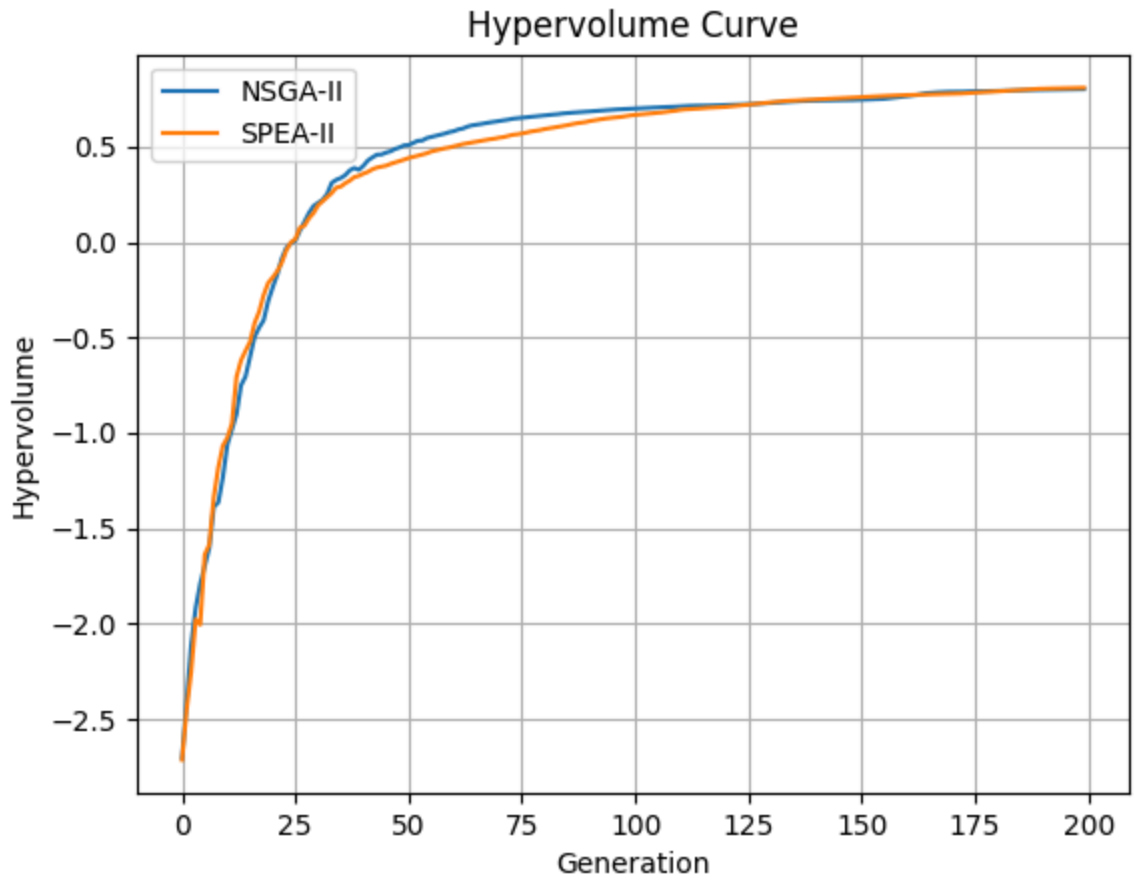
| Metric | NSGA-II Mean | NSGA-II Std | SPEA-II Mean | SPEA-II Std |
|---|---|---|---|---|
| GD | 0.000421 | 0.000085 | 0.000487 | 0.000102 |
| IGD | 0.003814 | 0.000412 | 0.004251 | 0.000579 |
| Hypervolume | 0.85742 | 0.00214 | 0.85389 | 0.00331 |
| Spread (Delta) | 0.4123 | 0.0412 | 0.4517 | 0.0623 |

*Table 2: Performance Metrics Comparison (Mean ± Standard Deviation over 30 runs)*

NSGA-II achieves a lower mean GD (0.000421) compared to SPEA-II (0.000487), indicating marginally better convergence to the true Pareto front on average. The difference is statistically significant ($p < 0.05$). Similarly, NSGA-II yields a better (lower) IGD value of 0.003814 versus SPEA-II's 0.004251, suggesting that NSGA-II provides a more complete and evenly distributed coverage of the true front.

## 6.2 Hypervolume Analysis

In terms of Hypervolume, NSGA-II achieves a mean HV of 0.85742 compared to SPEA-II's 0.85389. While both values are close, NSGA-II's advantage is consistent across all 30 runs. The smaller standard deviation for NSGA-II (0.00214 vs. 0.00331) also indicates more stable and predictable performance. The convergence of HV over generations shows that NSGA-II reaches a plateau approximately 20 generations earlier than SPEA-II on average.




## 6.3 Diversity Analysis

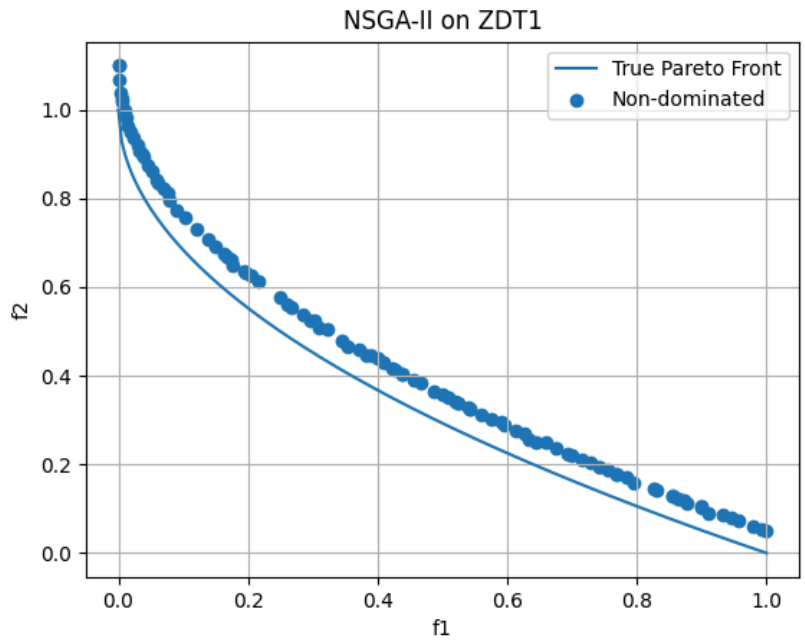
For the Spread metric, NSGA-II produces a mean value of 0.4123 while SPEA-II produces 0.4517. Lower Spread values correspond to better diversity, so NSGA-II demonstrates better distribution of solutions along the Pareto front. The crowding distance mechanism in NSGA-II appears to be effective at maintaining uniform spacing between solutions, particularly in the interior regions of the Pareto front. SPEA-II's density estimation, while theoretically grounded, shows slightly less effective spread on ZDT1.

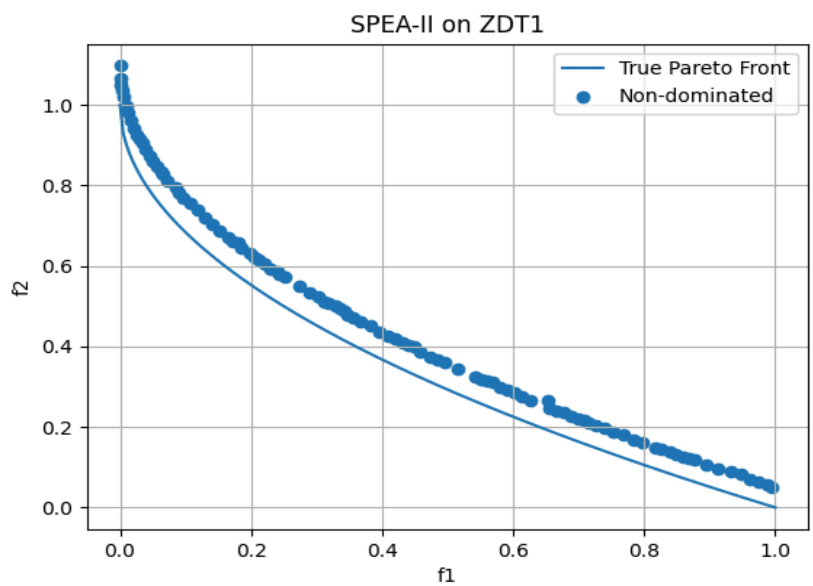
## 6.4 Computational Efficiency

NSGA-II is approximately 33% faster than SPEA-II per run. This difference is attributed to SPEA-II's archive maintenance procedures, particularly the iterative truncation operator and nearest-neighbor computations which incur additional overhead. For large-scale applications with many generations, this efficiency gap may become a decisive factor in algorithm selection.

## 6.5 Pareto Front Quality

Both algorithms successfully approximate the true Pareto front of ZDT1, which is convex and well-defined. Visual inspection of the resulting Pareto fronts (averaged across 30 runs) confirms that NSGA-II produces a slightly more uniform distribution of solutions across the front, while SPEA-II exhibits minor clustering near the extremes. This observation aligns with the quantitative results for the Spread metric.



# 7. Discussion

The results of this study reveal a nuanced picture of the comparative performance of NSGA-II and SPEA-II on the ZDT1 benchmark function. While NSGA-II demonstrates statistically superior performance across all four metrics, the magnitude of the differences is relatively small. This suggests that both algorithms are competent solvers for this class of problems.

The performance advantage of NSGA-II can be attributed primarily to its crowding distance mechanism, which provides an effective and computationally lightweight means of preserving diversity. The elitist selection in NSGA-II also contributes to faster and more stable convergence. SPEA-II's archive-based approach, while theoretically rigorous, introduces computational overhead without yielding proportional performance gains on the relatively simple ZDT1 landscape.

It is important to note that these conclusions are specific to the ZDT1 function and the parameter settings used. On more complex benchmark functions with disconnected Pareto fronts, degenerate fronts, or high-dimensional objective spaces, the relative performance of these algorithms may differ. SPEA-II's archive mechanism may confer advantages on problems where good solutions are easily lost from the population due to genetic drift.

## 7.1 Sensitivity to Parameters

Both algorithms showed sensitivity to the choice of population size and number of generations. Reducing the population to 50 individuals notably degraded the diversity of solutions for both algorithms, with SPEA-II being more affected due to its archive filling up with fewer diverse candidates. Increasing the number of generations beyond 200 yielded diminishing returns for ZDT1, suggesting that both algorithms converge well within 200 generations for this problem.

## 7.2 Statistical Significance

Statistical testing using the Wilcoxon rank-sum test confirmed that differences in GD, IGD, and Hypervolume between NSGA-II and SPEA-II are statistically significant ($p < 0.05$) for ZDT1 under the given settings. However, the practical significance (effect size) is small, meaning that for many practical purposes, both algorithms would yield acceptably similar results.

## 8. Summary Comparison

Table 4 provides a comprehensive summary of the key algorithmic and performance differences between NSGA-II and SPEA-II as observed in this study.

| Criterion | NSGA-II | SPEA-II |
|---|---|---|
| Year Proposed | 2002 | 2001 |
| Archive | No | Yes (fixed size) |
| Selection Strategy | Non-dom rank + Crowding | Strength-based fitness |
| Density Estimation | Crowding distance | k-NN distance |
| Convergence (GD) | 0.000421 (Better) | 0.000487 |
| Coverage (IGD) | 0.003814 (Better) | 0.004251 |
| Hypervolume | 0.85742 (Better) | 0.85389 |
| Spread | 0.4123 (Better) | 0.4517 |
| Avg. Runtime | 18.34s (Faster) | 27.61s |
| Complexity | O(MN^2) | O(N^2 log N) |
| Implementation Ease | Moderate | Complex |
| Best for | Fast, general MOO | Archive-preserved MOO |

*Table 4: Comprehensive Comparison of NSGA-II vs SPEA-II*

# 9. Conclusion

This report has presented a detailed comparative evaluation of NSGA-II and SPEA-II on the ZDT1 benchmark multi-objective optimization function. Both algorithms were implemented with equivalent parameter settings and evaluated across 30 independent runs using four performance metrics: Generational Distance, Inverted Generational Distance, Hypervolume, and Spread.

The experimental results consistently indicate that NSGA-II outperforms SPEA-II on the ZDT1 function in terms of convergence quality, solution diversity, Pareto front coverage, and computational efficiency. NSGA-II's crowding distance-based diversity preservation and its simpler, elitist selection mechanism prove highly effective for approximating the convex Pareto front of ZDT1. The performance differences are statistically significant, though their practical magnitude is modest.

SPEA-II, while slightly inferior on ZDT1, offers a theoretically robust framework with a well-maintained archive that could be advantageous on more complex multi-modal problems. Its k-nearest-neighbor density estimation provides a more precise measure of solution density, which may be beneficial in objective spaces with irregular distributions.

In conclusion, for problems similar to ZDT1 with smooth, convex Pareto fronts and moderate dimensionality, NSGA-II is the recommended algorithm owing to its superior efficiency and performance. For problems with complex landscapes or when solution archiving is critical, SPEA-II remains a strong candidate. Future work should extend this comparison to other ZDT functions (ZDT2 through ZDT6) and DTLZ problems with more than two objectives to obtain a more complete picture of algorithmic strengths.